



BASE24-atm[®]

EMEA AMEX Interface Manual

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Publications Number : BMSW-DS994-51

First Edition November 1998 /Release 5.0 /Version 1.0

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PREFACE

This document is primarily intended for users of the BASE24-atm EMEA AMEX Interface.

The BASE24-atm EMEA AMEX Interface is based on the BASE24 ISO Host Interface module (HISO) and is compatible with Release 5.0 and 5.1 of BASE24.

Intended Audience

Users should be familiar with the BASE24 configuration and Tandem systems. Users should also consult the relevant BASE24 or Tandem manuals where appropriate.

Location of Information

A list of section headings and related page references are given in the Contents page.

Section 1—Overview of the BASE24-atm EMEA AMEX Interface.

Section 2—Configuration Impact details changes to the BASE24 files to support the BASE24-atm EMEA AMEX Interface.

Section 3—File Access & Maintenance describes how to access details of the Application Files Terminal (AFT) screens for configuration of the BASE24-atm EMEA AMEX Interface.

Section 4—Message Types supported by the Interface, together with Transaction Message Flow diagrams and descriptions.

Section 5—Encryption for ATM Transactions details the security requirements for ATM transactions.

Appendix A—Response Code Translation defines the translation of response codes from AMEX to BASE24-atm.

Appendix B—Message Formats covers the updated external message formats.

Appendix C—Log Messages details log messages specific to the BASE24-atm EMEA AMEX Interface.

User's Comments Form

Related Documentation

BASE24 Documentation

BASE24 Log Message Reference Manual
Publications Number : BMBA-AE000-02

BASE24 Base Files Maintenance Manual
Publications Number : BMBA-AE000-03

BASE24 Text Command Reference Manual
Publications Number : BMBA-AE000-05

BASE24 LCONF Manual
Publications Number : BMBA-AE000-10

BASE24 Tokens Manual
Publications Number : BMBA-AE000-12

BASE24 Transaction Security Manual
Publications Number : BMBA-AE000-13

BASE24 ISO Host Interface Manual
Publications Number : BMBA-DH011-01

BASE24 External Messages Manual
Publications Number : BMBA-DH011-02

BASE24-atm Transaction Processing Manual
Publications Number : BMAT-AE000-07

XPNET Network Environment Source File Manual
Publications Number : BMXN-NT000-02

Third Party Documentation

AMEX - 3rd Party ATM Sharing Link - ISO 8583 General Requirements, Version 1.0, produced by AMEX, April 19 1995

User Comments

At the back of this manual, you will find a form that can be used to comment on any aspect of this document.

We hope that you will be able to help by telling us what you like and what improvements you would like to see in future editions.

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1. OVERVIEW

This document defines the BASE24-atm EMEA AMEX Interface. The functionality for the interface is based on the Release 5.0 BASE24 ISO Host Interface module (HISO).

The BASE24-atm EMEA AMEX Interface is compatible with Release 5.0 and Release 5.1 of BASE24. It supports ISO 8583 (1993) message formats and is Year 2000 compatible.

The BASE24-atm EMEA AMEX Interface supports Issuer-only capabilities, i.e. it supports BASE24 acquired transactions only.

Note that the character set supported by AMEX is EBCDIC, not ASCII.

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2. CONFIGURATION IMPACT

2.1 LOGICAL NETWORK CONFIGURATION FILE (LCONF)

The Logical Network Configuration File (LCONF) is the central location of file information and parameters for BASE24. There is one LCONF for each Logical Network.

The LCONF contains *assigns* which link the physical names of objects in the logical network to names used in the BASE24 code, and *params* which supply parameters and processing values to BASE24.

For more details of the standard LCONF assigns and parameters, please refer to the BASE24 LCONF Manual.

2.1.1 Assigns

EMF

The fully qualified file name of the External Message File (EMF).

Example: \B24.\$DATA.PRO1DATA.EMF

HCF

The fully qualified file name of the Host Configuration File (HCF).

Example: \B24.\$DATA.PRO1DATA.HCF

KEYF

The fully qualified file name of the Key File (KEYF).

Example: \B24.\$DATA.PRO1DATA.KEYF

LMCF

The fully qualified file name of the Log Message Configuration File (LMCF). The LMCF assign is used by all of the application processes and, optionally, any XPNET process that can alter log messages.

Example: \B24.\$DATA.PRODCNTL.LMCF

PLF

Identifies the name of the PCS Log File (PLF). This assign is used to monitor the network.

Example: \B24.\$DATA.PRO1DATA.PLF

SAF

The fully qualified file name of a Store-and-Forward File (SAF).

Examples: \B24.\$DATA.PRO1DATA.SAF1 (data file name)

 \B24.\$DATA.PRO1TPLT.SAF (template file name)

2.1.2 Parameters

LOGICAL-NET

The LOGICAL-NET param contains one to four alphanumeric characters identifying the valid name of the logical network.

If the param is not present in the LCONF or contains an invalid value, the processes that access the LCONF for this param and the processes that contain objects that access the LCONF for this param abend during initialisation.

Param Default: No default value

SAF-NOTIFY-INTERVAL

The number of writes made to the Store-and-Forward File (SAF) between calculating the percentage of file full.

Param Default: 50

SAF-NOTIFY-PERCENTAGE

The percentage of file full at which the ISO Host Interface process logs a message indicating the Store-and-Forward File (SAF) is becoming full.

Valid values are 10 through 99.

Param Default: 80

ATM-FORWARD-INST-ID

Required to handle field 94 in Network Management Messages (NMMs) processing.

Format: Up to 11 numeric characters.

ATM-AMEX-ISSUER-INST-ID

Required to handle field 93 in Network Management Messages (NMMs) processing.

Format: Up to 11 numeric characters.

ATM-LN-CUTOVER

Specifies the logical network cutover time for BASE24-atm. Mandatory.

Format: ATM-LN-CUTOVER

ATM-AMEX-ACQ-INST-ID-nn

where nn is a number between 01 and 99 indicating the Terminal Owner FIID.

Required to handle fields P32 and P58 which, for AMEX, are handled differently to standard ISO processing. Instead of the field being generated by BASE24, this field must come from AMEX. As there may be a number of acquirers with which BASE24 communicates, the parameter must be processed with the Terminal Owner ID as part of the processing.

The Terminal Owner FIID is a field up to 4 alphanumeric characters. The Acquirer Institution ID is a numeric field of between 1 and 11 characters. A semi-colon separates the two fields. A 'catchall' may also be entered into the parameter which has the following attribute ****;ACQ-INST-ID. Appropriate log messages are output for any field that is entered incorrectly.

A global table is created for the number of entries found by a search of all LCONF parameters with any number between 01 and 99. This table is built using the Terminal Owner FIID and the Acquirer Institution ID from the entered parameters. If the 'catchall' parameter is found, the Acquirer Institution ID found there should not be included in the global table, but should be put into a global (which has been initialised to spaces).

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3. FILE ACCESS & MAINTENANCE

This section provides descriptions of the BASE24 screens which have enhanced usage with the BASE24-atm EMEA AMEX Interface. The screen descriptions provide details of the fields on the screen including valid field values, default values and field length.

3.1 SCREEN LAYOUT & KEYBOARD CONVENTIONS

3.1.1 Standard BASE24 Screen Layout

File maintenance screens display the following header line at the top of each screen display:

BASE24-<prod> <type> LLLL FIID YY/MM/DD HH:MM nn OF nn

where:

<prod>	specifies the screens as part of the BASE24 system for the product specified.
<type>	specifies the type of screen displayed.
LLLL	specifies the Logical Network and displays the Logical Network Identifier entered at individual operator Logon.
FIID	specifies the identification of the financial institution.
YY/MM/DD	specifies the current system processing date. This field is displayed whenever an AFT screen is accessed. The format of the field is configurable according to the LCONF parameter, DATE-DISPLAY-FORMAT.
HH:MM	specifies the current system processing time at the time the screen was displayed. This field is displayed whenever an AFT screen is accessed.
nn OF nn	specifies the screen number of the screen currently displayed as well as the total number of screens for that particular file.

3.1.2 Standard BASE24 Function Keys

The standard BASE24 function keys are used on BASE24 screens. When the usage of these function keys varies with BASE24 files and functions, the differences are noted with the screen descriptions for the file.

BASE24 has an online Help screen which lists the function keys that apply on each screen. The Help screen is displayed by pressing the F12 function key on Tandem 6520 and 6530 terminals and the PF12 function key on IBM 3270 terminals.

The usage of the function keys displayed on the screen is explained in the following table. The first column shows the function key on Tandem 6520 and 6530 terminals. The second column shows the function key on IBM 3270 terminals. The third and fourth columns provide a description of how the function key works for the screen.

TANDEM	IBM	FUNCTION	DESCRIPTION
F1	ENTER	VALIDATE DATA	Checks that the entered data conforms to the specified standards or values.
F2	PF2	READ RECORD	Reads a record from the current file.
F3	PF3	ADD RECORD	Adds a unique record to the current file.
F4	PF4	DELETE RECORD	Deletes a record from the current file.
F5	PF5	UPDATE RECORD	Changes an existing record in the current file.
F6	PF6	READ NEXT RECORD	Reads the next record in the file in which the user is working.
F7	PF7	GOTO NEW PAGE	Displays a different screen in the current file.
F8	PF8	CLEAR SCREEN	Clears any values in the record to spaces or replaces them with default values. This impacts all screens associated with the file accessed including the current screen, any previous screens, and any remaining screens. Once this function has been used, the update function cannot be completed until the read function has been completed.
F9	PF9	NEXT SCREEN	Displays the next screen of the file accessed. If the last screen was displayed and this function key was pressed, the first screen of the file will be displayed.
F10	PF10	PRINT SCREEN	Prints a hard copy of the displayed screen. If a printer is not specified and the system default is used, the print job will remain in the spooler until the location/status is changed by the operator.
F11	PF11	PREVIOUS SCREEN	Displays the previous screen of the file accessed. If the first screen was displayed and this function key was pressed, the last screen of the file will be displayed.
F12	PF12	DISPLAY HELP SCREEN	Displays details on a command or task (STANDARD HELP SCREEN ONLY).
F13	PF13	CHANGE CURRENT LOGICAL NET	Changes the current logical network while in a file.

TANDEM	IBM	FUNCTION	DESCRIPTION
F16	PA1	EXIT OR GOTO FILE	Exits BASE24 or the file currently accessed.
SF16	PA2	LOGOFF	Ends a session as a terminal or operator.

3.2 USING ONLINE HELP

All information displayed on the Help Screens below is read only.

Press any key with the exception of SF9 - SF16, F10, and F16 to display the screen from which the F12 (HELP) function key was pressed.

3.2.1 Example of Help Screens

Figure 3-1 is an example of a BASE24 Help screen.

```

BASE24-BASE KEY FILE DEM1 94/11/28 10:10 01 OF 03
DPC/FIID: INTERFACE PROCESS:

STANDARD FILE MAINTENANCE FUNCTION FOR KEYS 6520/6530 (3270)
F1 (ENTER) - VALIDATE DATA
F2 (PF2) - READ RECORD
F3 (PF3) - ADD RECORD
F4 (PF4) - DELETE RECORD
F5 (PF5) - UPDATE RECORD
F6 (PF6) - READ NEXT RECORD
F7 (PF7) - GOTO NEW PAGE
F8 (PF8) - CLEAR SCREEN
F9 (PF9) - NEXT SCREEN
F10 (PF10) - PRINT SCREEN
F11 (PF11) - PREVIOUS SCREEN
F12 (PF12) - DISPLAY HELP SCREEN
F13 (PF13) - CHANGE CURRENT LOGICAL NET
F16 (PF16) - EXIT OR GOTO FILE
SF16 (PA2) - LOGOFF

***** BASE24 *****
NEW PAGE: █ FILE DESTINATION: NEW LOGICAL NETWORK ID:
ANY FUNCTION KEY EXCEPT SF9-SF16, F10, OR F16 RETURNS. F10 PRINTS AND RETURNS

```

Figure 3-1 Example of a BASE24 Help Screen

3.3 FILE ACCESS

All the BASE24 product files and functions that can be accessed are listed on the BASE24 Product Menu, accessed from the Virtual Menu.

Only the files and screen groupings available for a particular logon will be displayed for that logon.

3.3.1 BASE24 Virtual Menu

Figure 3-2 is an example of a Virtual Menu screen.

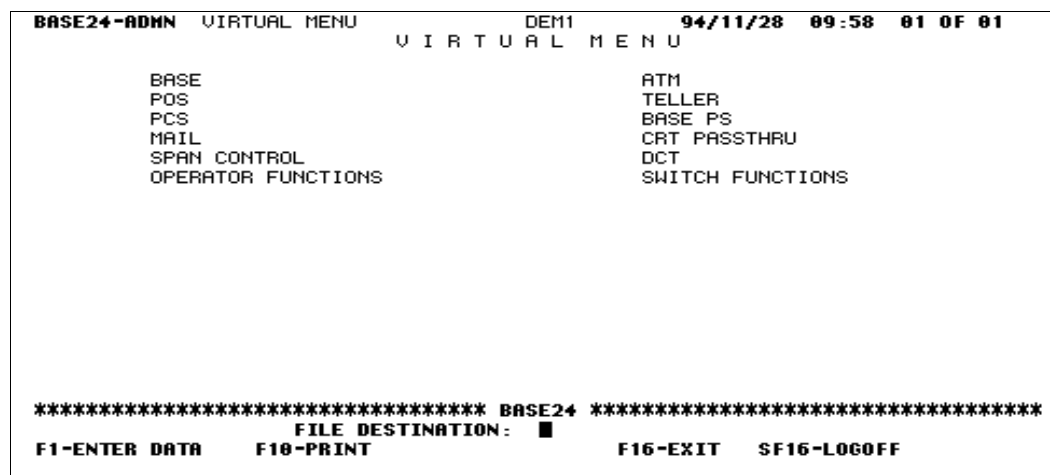


Figure 3-2 Example of a Virtual Menu

3.3.2 BASE24 Product Menu

To display a particular BASE24 Product Menu screen, select the option by placing the cursor on the required item on the Virtual Menu and press the F1 function key. The specified BASE24 Product Menu screen will be displayed.

Below are examples of BASE24 Product Menu screens.

BASE24-ADMN MENU		DEM1	94/11/28 10:00 01 OF 01
B A S E P R O D U C T M E N U			
ARF	CAF	CPF	ECF
EMF	HCF	IDF	KEVA
KEYF	NCSP	NEG	PBF
PRE	SEC	SPF	TKN
UAF			
***** BASE24 *****			
FILE DESTINATION: █			
F1-ENTER DATA	F10-PRINT	F12-HELP	F16-EXIT SF16-LOGOFF

Figure 3-3 Example of the BASE24 Product Menu

3.4 HOST CONFIGURATION FILE (HCF)

This section provide a description of the Host Configuration File (HCF) screens which are used with the BASE24-atm EMEA AMEX Interface. The HCF contains one record for each unique Data Processing Centre (DPC) and BASE24-atm EMEA AMEX Interface process pair in the logical network.

If both the BASE24-atm EMEA AMEX Interface module and the BASE24-pos EMEA AMEX Interface module are installed, separate HCF records must be configured for ATM and POS traffic.

Each HCF record contains pertinent information regarding host message formats, timer limits, and request and response sequences.

The HCF is secured under Tandem group level security so that only authorised network operators can access or start programs that access the file.

The HCF defines BASE24-to-host communications links for transaction message traffic routed to a host authorisation system, e.g. AMEX. Its co-ordination of BASE24-to-host data communications activity ensures that the host receives requests and BASE24 processes respond properly to the requests. The key to the records in the HCF is the DPC number and the AMEX Interface process name.

For full details of the standard usage of the HCF, refer to the BASE24 Base Files Maintenance Manual.

3.4.1 HCF Screen 1

HCF screen 1 (Figure 3-4) displays timer limits, processing flags, and data prefix characters. HCF screen 1 is shown below, followed by descriptions of its fields.

BASE24-BASE HOST CONFIGURATIONLLLL YY/MM/DD HH:MM 01 OF 25
DPC NUMBER: 0000 HISF NAME: P1A^HISF1

TIMER LIMITS

TOKEN GROUP: _____
PROCESSING FLAGS

NETWORK MANAGEMENT: _____ (SEC)
EXTENDED NETWORK: _____ (SEC)
STORE AND FORWARD: _____ (SEC)
WAIT FOR TRAFFIC: _____ (SEC)
PERFORMANCE PERIOD: _____ (MIN)

MAXIMUM OUTSTANDING SAFS: _____
MAXIMUM OUTSTANDING REQUESTS
OUTBOUND: _____ INBOUND: _____

RELEASE INDICATOR: _____

STORE AND FORWARD: _____
DPC TYPE: _____
ACK TO DPC: _____ (Y/N)
ACK FROM DPC: _____ (Y/N)
MAX TIMEOUTS: _____
NMM ENABLED: _____ (Y/N)
MAX SAF RETRY: _____
PROTOCOL TYPE: _____
MESSAGE SEQUENCE FLAG: _____
TIME DISCREPANCY CHECK: _____ (Y/N)
MESSAGE FORMAT: _____

DATA PREFIX CHARACTERS

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

VALID ENTRIES FOR DATA PREFIX CHARACTERS ARE 0 THRU 9 AND A THRU F.

BASE24

NEW PAGE:

FILE DESTINATION:
F12-HELP

NEW LOGICAL NETWORK ID:

Figure 3-4 HCF Screen 1

DPC NUMBER

4 (N) M

The number of the institution’s Data Processing Centre (DPC). This is the host computer that processes messages received from a BASE24 product. The default must be changed to reflect the AMEX host. The values in this field and the HISF NAME field combine to make the primary key. Valid values are 0001 to 9999.

Default Value: This field has a default value of 0000. However, this value must be changed to a valid value.

HISF NAME

16 (AN) M

The symbolic name of the BASE24-atm EMEA AMEX Interface process that communicates with the specified DPC, as defined in the Network Environment Source File (NEFS).

Default Value: P1A^HISF1

TOKEN GROUP**4 (AN)**

Not used by the BASE24-atm EMEA AMEX Interface.

TIMER LIMITS

The following fields are used to configure timer limits that are not product-specific:

NETWORK MANAGEMENT

EXTENDED NETWORK

STORE AND FORWARD

WAIT FOR TRAFFIC

PERFORMANCE PERIOD

These fields appear in a column on the screen. However, field descriptions are arranged according to cursor movement (that is, row first instead of column first).

Therefore, descriptions for fields in the first column (timer limits) and descriptions for fields in the second column (processing flags) are combined.

PROCESSING FLAGS

All fields on HCF screen 1 except the DPC NUMBER, HISF NAME, TOKEN GROUP, and timer limits are used to configure the processing options that are not product-specific.

NETWORK MANAGEMENT**4 (N) M**

The number of seconds that a BASE24 product waits for a response after transmitting a network management message to a DPC station. Network management messages include echo-test, logon, logoff, and dynamic key management messages.

If a response message is not received within the time interval specified in this field, the Host Interface process marks the station as down. Valid values are 0 to 9999.

Default Value: 30

STORE AND FORWARD**1 (N) M**

A code specifying the method the AMEX Interface process uses for sending store-and-forward messages to the DPC. Store-and-forward processing is invoked whenever a DPC is available and there are store-and-forward messages to be sent.

Valid values are as follows:

- 0 Intersperse store-and-forward messages with real-time messages going to a host for approval. Under this method, real-time transactions are processed normally, with the exception of reversals, which are placed in the Store-and-Forward File (SAF) in case the message being reversed has not yet been processed. (Default)
- 1 Transmit all store-and-forward messages prior to any real-time transactions. Under this method, if there are any messages for a DPC in the Store-and-Forward File (SAF), these messages are sent prior to any real-time transactions.

EXTENDED NETWORK**4 (N) M**

The number of seconds to set the extended network management message timer when it is used. Valid values are 0 to 9999.

When the Host Interface process marks a station down, it sets a timer using the number of seconds specified here. If no subsequent messages are received from the station prior to the expiration of the timer, the Host Interface process sends an echo-test message.

The value entered in this field is also the number of seconds the AMEX Interface process is to wait, once the initial message has timed out, before resending dynamic key management messages (that is, new key and change key messages).

Default Value: 60

DPC TYPE**1 (N) M**

A code specifying whether the DPC's stations for an issuer host are single-threaded or multithreaded. Acquirer hosts can be either regardless of the value in this field. Valid values are as follows:

- 0 Multithreaded. Multiple messages can be outstanding to a single station at one time. A response is not required to each message before another message is sent to the station. (Default)
- 1 Single-threaded. Only one message can be outstanding to a station at one time. A response is required to each message before another message can be sent to the station.

STORE AND FORWARD**4 (N) M**

The number of seconds the AMEX Interface process waits for an acknowledgment after transmitting a store-and-forward message to the DPC. Valid values are 0 to 9999.

Default Value: 30

ACK TO DPC**1 (AN) M**

A code indicating whether the AMEX Interface process sends text-level acknowledgments to the DPC. For BASE24-atm EMEA AMEX Interface processing set to the default value of 'N'. Valid values are as follows:

- Y Yes, send text-level acknowledgments.
- N No, do not send text-level acknowledgments (default).

WAIT FOR TRAFFIC**4 (N) M**

The number of seconds the AMEX Interface process waits for traffic on the line to a station before sending an echo-test message. Valid values are 0 to 9999.

Default Value: 60

ACK FROM DPC**1 (AN) M**

A code indicating whether the AMEX Interface process expects text-level acknowledgments from the DPC. For BASE24-atm EMEA AMEX Interface processing set to a value of 'Y'. Valid values are:

- Y Yes, expect text-level acknowledgments.
- N No, do not expect text-level acknowledgments (default).

PERFORMANCE PERIOD**4 (N) M**

The number of minutes the AMEX Interface process uses as an interval for tracking performance of the DPC. BASE24 products enable network operators to request performance statistics via a text command. The value in this field specifies the number of minutes the AMEX Interface process uses as an interval for tracking DPC performance.

Default Value: 20

MAX TIMEOUTS**2 (N) M**

The maximum consecutive number of non-network management messages that can time-out before the AMEX Interface process marks the station down. A time-out on an 0800 message causes the station to be marked down immediately, without regard for this maximum. Valid values are 1 to 99.

Default Value: 2

NMM ENABLED**1 (AN) M**

A code specifying whether the AMEX Interface process can initiate network management messages (NMMs) to the DPC. Valid values are:

Y Yes, the AMEX Interface process can initiate NMMs to the DPC (default).

N No, the AMEX Interface process cannot initiate NMMs to the DPC.

This value does not apply to key management messages. Key management messages are controlled in the Key File (KEYF), detailed in section 3.5.

MAXIMUM OUTSTANDING SAFS**4 (N) M**

The maximum number of store-and-forward messages that can be outstanding to a multithreaded host at one time. The AMEX Interface process uses the value in this field when the DPC TYPE field contains a 0, identifying a multithreaded host. Valid values are 1 through 9999.

The only types of messages that get placed in the Store-and-Forward File (SAF) are advices and reversals. NMMs, including key management messages, are never placed in the SAF.

Default Value: 1

MAX SAF RETRY**2 (N) M**

The number of times the AMEX Interface process should attempt to send the same store-and-forward message before deleting it from the file. This option can be used to circumvent processing when it becomes apparent that the message itself is probably at fault for the DPC's inability to process it.

Once a store-and-forward message has been sent this number of times, the AMEX Interface process writes the message to its log, deletes the message from the Store-and-Forward File (SAF), and continues its store-and-forward processing with the next message in the SAF. Valid values are as follows:

Set this value to 2 for BASE24-atm EMEA AMEX Interface processing.

MAXIMUM OUTSTANDING REQUESTS

The values in the OUTBOUND and INBOUND fields specify how many outstanding requests can be queued outbound to a host and inbound from a host. If the maximum outbound to a host is reached and the BASE24 product is configured to do so, authorisation is done by the BASE24 product in lieu of the host.

0 Do not limit the number of times a store-and-forward message can be sent.

1–99 Limit sending store-and-forward messages to the number of times indicated.

OUTBOUND**4 (N) M**

The maximum number of outstanding requests that can be queued to a host. When the maximum is reached, the excess requests are returned to the originating process with a response indicating that the host is unavailable.

Default Value: 30

INBOUND**4 (N) M**

Not used by the BASE24-atm EMEA AMEX Interface.

PROTOCOL TYPE**2 (N) M**

A value indicating the type of protocol message header processing to be performed by the AMEX Interface process. Valid values are as follows:

- 00 No special protocol processing required. Message header processing is performed by the XPNET process (default).
- 01 SNA/CICS protocol.
- 02 Bisync protocol.
- 03 UK1 X.25 protocol (this value is not used).

MESSAGE SEQUENCE FLAG**1 (N) M**

A code indicating whether message sequencing should be done at the process level (standard) or at the station level. Valid values are as follows:

- 0 AMEX Interface process level (default)
- 1 Station level (reserved for future use)

TIME DISCREPANCY CHECK**1 (AN) M**

A code indicating whether a time discrepancy check is made for incoming 0800 messages. If a difference of five minutes or more exists between the time the 0800 message is sent and the time the message is received by the AMEX Interface, the AMEX Interface generates a warning message. Valid values are as follows:

- Y Yes, a time discrepancy check is made.
- N No, a time discrepancy check is not made (default).

RELEASE INDICATOR**2 (N) M**

A code specifying the release of BASE24 ISO-based network management messages being exchanged with this DPC. The valid value is:

01 Current release (Release 5.0 and Release 5.1) - Default

MESSAGE FORMAT**2 (N) M**

A code specifying whether certain data elements in the BASE24 ISO-based network management messages are in fixed or variable format. Valid values are as follows:

00 Fixed format

01 Variable format (default)

DATA PREFIX CHARACTERS**9 fields of 2 hexadecimal characters each**

The nine fields that follow on the screen allow entry of ASCII data prefix characters. The data prefix characters are placed at the front of all BASE24 messages to the DPC.

Each time a message is sent to the DPC, the AMEX Interface process prefixes the message with the data prefix characters specified here. Up to nine characters, stored in hexadecimal character display format, can be placed in front of the messages sent to the DPC. For example, 41 is the hexadecimal representation of the ASCII letter A.

Since these are representations of ASCII, not EBCDIC, codes, DPCs connected to EBCDIC lines translate these codes to EBCDIC along with the rest of the message when transmitting the message.

The characters entered in these fields must be left-justified. If nothing is entered in the field, no data prefix characters will be included in the message header.

Valid values are hexadecimal characters 00 through FF. However, the following hexadecimal characters should not be included since they can represent protocol characters depending on the protocol: 01 to 06, 10, 15, 16, 17, and 1F. BASE24 products do not edit for these values.

Default Value: No default value.

3.4.2 HCF Screen 5

HCF screen 5 (Figure 3-5) contains processing parameters and timer limits for BASE24-atm messages. HCF screen 5 is shown below, followed by descriptions of its fields.

```

BASE24-ATM HOST CONFIGURATION      LLLL          YY/MM/DD HH:MM 05 OF 25
                                DPC NUMBER: 0000      HISF NAME: P1A^HISF1

                                ATM PRODUCT DATA

                                AUTH PROCESS: _____
                                MESSAGE FORMAT: _____
                                RELEASE INDICATOR: _____

                                TIMER LIMITS

                                OUTBOUND LIMIT: _____ (SEC)
                                INBOUND LIMIT: _____ (SEC)
                                COMPLETION: _____ (SEC)
                                COMPLETION ACK: _____ (SEC)
                                QUEUE SUBTRACT: _____ (SEC)

*****
*****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                                F12-HELP
                                BASE24

```

Figure 3-5 HCF Screen 7

ATM PRODUCT DATA

The following fields contain processing parameters and timer limits for BASE24-atm messages.

AUTH PROCESS

16 (AN)

Not used by the BASE24-atm EMEA AMEX Interface.

MESSAGE FORMAT

2 (N) M

A code specifying whether certain data elements in the BASE24 ISO-based messages are in fixed or variable format. The valid value for the BASE24-atm EMEA AMEX Interface is:

01 Variable format (default)

RELEASE INDICATOR**2 (N) M**

A code specifying the release of BASE24 ISO-based messages being exchanged with this DPC. The valid value for the BASE24-atm EMEA AMEX Interface is:

01 Current release (Release 5.0 and Release 5.1) - Default

TIMER LIMITS

The following fields contain timer limits for processing BASE24-atm messages. Embedded blanks are not allowed in these fields.

Note:

These timer limits apply to BASE24-atm messages other than network management messages. Network management message timers are set on HCF screen 1.

OUTBOUND LIMIT**4 (N) M**

The number of seconds that the AMEX Interface process waits for a response message after transmitting a request message to the host. Valid values are 0 to 9999.

Default Value: 15

INBOUND LIMIT**4 (N) M**

Not used by the BASE24-atm EMEA AMEX Interface.

COMPLETION**4 (N) M**

The time limit in seconds associated with a completion message in either direction for an ATM transaction. The value in this field is not used in processing.

Default Value: 30

COMPLETION ACK**4 (N) M**

Not used by the BASE24-atm EMEA AMEX Interface.

QUEUE SUBTRACT**4 (N) M**

Not used by the BASE24-atm EMEA AMEX Interface.

3.5 KEYF FILE

The Key File (KEYF) contains the information and parameters required by the BASE24-atm EMEA AMEX Interface for PIN encryption and PIN translation.

It contains one record for each DPC number and AMEX Interface process combination defined in the Host Configuration File (HCF), allowing for individual control of the above-mentioned processing for each of these entities.

The following screens are used to access records in the KEYF for the BASE24-atm EMEA AMEX Interface:

- Screen 1—contains PIN encryption and PIN translation details. Message authentication parameters, intermediate keys, PIN key exchange keys and Message Authentication Code (MAC) key exchange keys are **not** used by the BASE24-atm EMEA AMEX Interface.
- Screen 2—contains inbound and outbound key information for PINs. MAC key fields are not used by the BASE24-atm EMEA AMEX Interface.
- Screen 3—not used.

3.5.1 KEYF Screen 1

KEYF screen 1 contains PIN encryption, PIN translation, and message authentication parameters, intermediate keys, PIN key exchange keys and MAC key exchange keys.

KEYF screen 1 is shown below, followed by descriptions of its fields.

BASE24-BASE KEY FILE		LLLL	YY/MM/DD HH:MM 01 OF 03
DPC/FIID:		INTERFACE PROCESS:	
ENCRYPT TYPE:	_____	BASE24 ENCRYPT TYPE:	_____
PIN BLOCK FORMAT:	_____	ANSI PAN FORMAT:	_____
MAC ENCRYPT TYPE:	_____	PIN PAD CHARACTER:	_____
MAC DATA TYPE:	_____	NUMBER OF KEYS:	_____
KEY LENGTH:	_____	FULL MESSAGE MAC:	_____
INTERMEDIATE KEYS			
CLEAR:	_____	CHECK DIGITS:	_____
ENCRYPTED:	_____		
EXCHANGE KEYS			
PIN KEY:	_____	CHECK DIGITS:	_____
MAC KEY:	_____	CHECK DIGITS:	_____
*****			BASE24

NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:	
	F12-HELP		

Figure 3-6 KEYF Screen 1

DPC/FIID

4 (AN) M

The DPC number of the host or FIID of the interchange using this KEYF record. The value in this field matches the value in the DPC NUMBER field on Host Configuration File (HCF) screen 1.

Default Value: No default value.

INTERFACE PROCESS

16 (AN) M

The name of the interface process associated with the DPC or interchange identified in the DPC/FIID field. The value in this field matches the value in the HISF NAME field on Host Configuration File (HCF) screen 1.

Default Value: No default value.

ENCRYPT TYPE**1 (N) M**

Specifies the type of PIN encryption used or expected by the DPC.

- 0 Clear PINs (default)
- 1 Security module PIN encryption
- 2 Software DES PIN encryption

BASE24 ENCRYPT TYPE**1 (N) M**

Specifies the type of PIN management used by BASE24 products. Valid values are as follows:

- 0 Clear PINs (default)
- 1 Security module PIN management

PIN BLOCK FORMAT**1 (N) M**

Specifies the PIN block format of the PIN in the inbound and outbound external messages. Valid values are as follows:

- 0 Clear PINs (default)
- 1 ANSI (PIN/PAN) PIN block
- 3 PIN/PAD PIN block

Note: Although value 1 is called the ANSI PIN block, it is also known as the PIN/PAN PIN block because it includes three PAN formats, only one of which is part of the ANSI standard. The PAN format being used is specified in the ANSI PAN FORMAT field on this screen.

ANSI PAN FORMAT**1 (N) M**

If the value in the PIN BLOCK FORMAT field is 1, the value in this field specifies which PAN digits are used in the formation of the external message's PIN/PAN PIN block. Valid values are as follows:

- 0 12 right-most digits of the PAN, excluding the check digit (ANSI standard - default)
- 1 12 right-most digits of the PAN, including the check digit
- 2 12 left-most digits of the PAN

MAC ENCRYPT TYPE**1 (N) M**

Not used by the BASE24-atm EMEA AMEX Interface.

PIN PAD CHARACTER**1 (AN) M**

If the value in the PIN BLOCK FORMAT field is 3 (PIN/PAD), the value in this field specifies the PAD character used in the formation of the external message's PIN block. Valid values are A through F.

Default Value: F

MAC DATA TYPE**1 (N) M**

A code specifying the character set in which messages are being transmitted between the DPC or interchange and the interface. Valid values are:

- 0 ASCII (default)
- 1 EBCDIC

Not used by the BASE24-atm EMEA AMEX Interface.

NUMBER OF KEYS**1 (N) M**

A code identifying whether the inbound and outbound keys are combined or separate. Valid values are as follows:

- 1 Combined. The inbound and outbound PIN keys are the same and the inbound and outbound MAC keys are the same. (Default.)
- 2 Separate. The inbound and outbound PIN keys are different and the inbound and outbound MAC keys are different.

Not used by the BASE24-atm EMEA AMEX Interface.

KEY LENGTH**1 (N) M**

A code identifying whether single- or double-length Key Exchange Keys (KEKs) are being used with this interface. Valid values are:

- 1 Single-length KEKs (default)
- 2 Double-length KEKs

Not used by the BASE24-atm EMEA AMEX Interface.

FULL MESSAGE MAC**1 (AN) M**

A code specifying whether selected data elements or the entire message is considered when computing the MAC. Valid values are:

- Y Yes, the full message is authenticated
- N No, the full message is not authenticated (default)

Not used by the BASE24-atm EMEA AMEX Interface.

INTERMEDIATE KEYS

The Intermediate Key fields are not used by the BASE24-atm EMEA AMEX Interface. The fields all default to a hexadecimal value of zeros.

EXCHANGE KEYS

The Exchange Key fields are not used by the BASE24-atm EMEA AMEX Interface. The fields all default to a hexadecimal value of zeros.

3.5.2 KEYF Screen 2

KEYF screen 2 consists of inbound and outbound working keys for PINs and MACs. KEYF screen 2 is shown below, followed by descriptions of its fields.

BASE24-BASE KEY FILE
DPC/FIID:

LLLL
INTERFACE PROCESS:

YY/MM/DD HH:MM
02 OF 03

OUTBOUND KEYS:

PIN KEY1:

PIN KEY2:

INBOUND KEYS:

PIN KEY1:

PIN KEY2:

PIN KEY INFORMATION

CHECK DIGITS1:

CHECK DIGITS2:

CHECK DIGITS1:

CHECK DIGITS2:

CURRENT INDEX:

KEY COUNTER:

CURRENT INDEX:

KEY COUNTER:

OUTBOUND KEYS:

MAC KEY1:

MAC KEY2:

INBOUND KEYS:

MAC KEY1:

MAC KEY2:

MAC KEY INFORMATION

CHECK DIGITS1:

CHECK DIGITS2:

CHECK DIGITS1:

CHECK DIGITS2:

CURRENT INDEX:

KEY COUNTER:

CURRENT INDEX:

KEY COUNTER:

NEW PAGE:

FILE DESTINATION:

NETWORK ID:

F12-HELP

BASE24

NEW LOGICAL

Figure 3-7 KEYF Screen 2

PIN KEY INFORMATION

The fields on the upper part of this screen contain working keys for PINs. The fields contain the PIN keys used for outbound transactions (used by the BASE24-atm AMEX interface in messages sent to AMEX).

If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1

16 hexadecimal characters M

A key used for the encryption or translation of PINs. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Default Value: 0000000000000000

CHECK DIGITS1**4 hexadecimal characters M**

The check digits corresponding to the value in the PIN KEY1 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

Default Value: 0000

CURRENT INDEX**1 (N) M**

Indicates which outbound or inbound PIN encryption key is currently being used by the interface. Valid values are:

- 1 Value in the PIN KEY1 field (default)
- 2 Value in the PIN KEY2 field

PIN KEY2**16 hexadecimal characters M**

A key used for the encryption or translation of PINs. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Default Value: 0000000000000000

CHECK DIGITS2**4 hexadecimal characters M**

The check digits corresponding to the value in the PIN KEY2 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

Default Value: 0000

KEY COUNTER**6 hexadecimal characters M**

The number of times the outbound or inbound PIN key has been changed by dynamic key management processing.

Not used by the BASE24-atm EMEA AMEX Interface.

MAC KEY INFORMATION

The fields on the lower part of this screen contain working keys for MACs.

The MAC Key fields are not used by the BASE24-atm EMEA AMEX Interface. The fields all default to a hexadecimal value of zeros.

4. MESSAGE TYPES

The BASE24-atm EMEA AMEX Interface module supports issuer processing. It also supports network management message types.

The following table indicates the message types the BASE24-atm EMEA AMEX Interface supports.

Type	Description	Product
1200	Financial transaction request	ATM
1210	Financial transaction response	ATM
1420	Reversal request	ATM
1430	Reversal response	ATM
1804	Network Management request	BASE
1814	Network Management response	BASE

4.1 TRANSACTION MESSAGE FLOWS

This section describes the message flows for the following transactions sent from BASE24 and responded to by AMEX.

- Requests
- Reversals

4.1.1 Financial Transaction Request/Response Message (1200/1210)

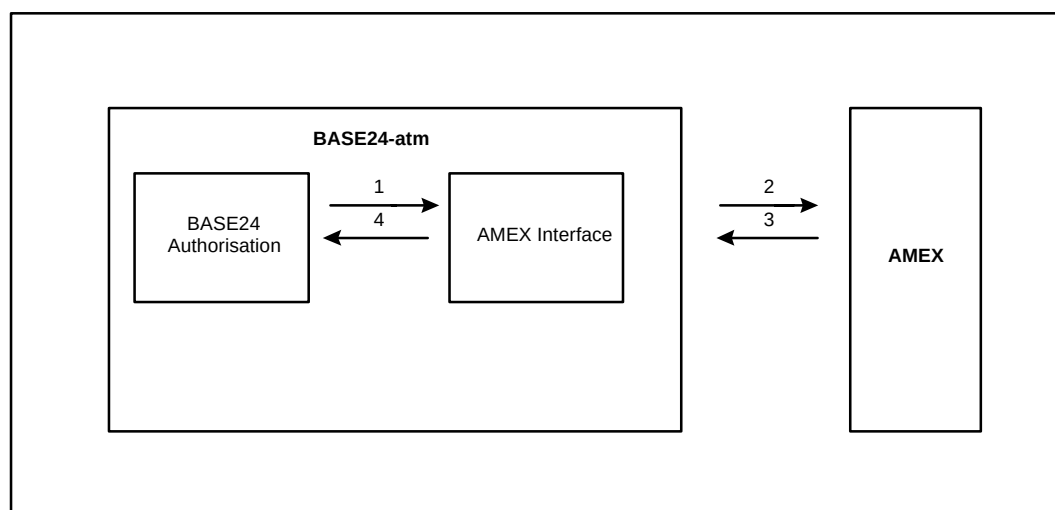


Figure 4-1 Financial Transaction Request/Response Message

1. A BASE24 Authorisation process sends the request to the BASE24-atm AMEX Interface in the 0200 internal message format.
2. The BASE24-atm AMEX Interface reformats the message into the 1200 external message format and sends the external message to AMEX for authorisation.
3. AMEX approve or decline the transaction, format the response into the 1210 external message format and send it to the BASE24-atm AMEX Interface.
4. The BASE24-atm AMEX Interface formats an 0210 internal response message and sends the response to the BASE24 Authorisation process to be delivered to the ATM.

4.1.2 Financial Transaction Reversal Request/Response Message (1420/1430)

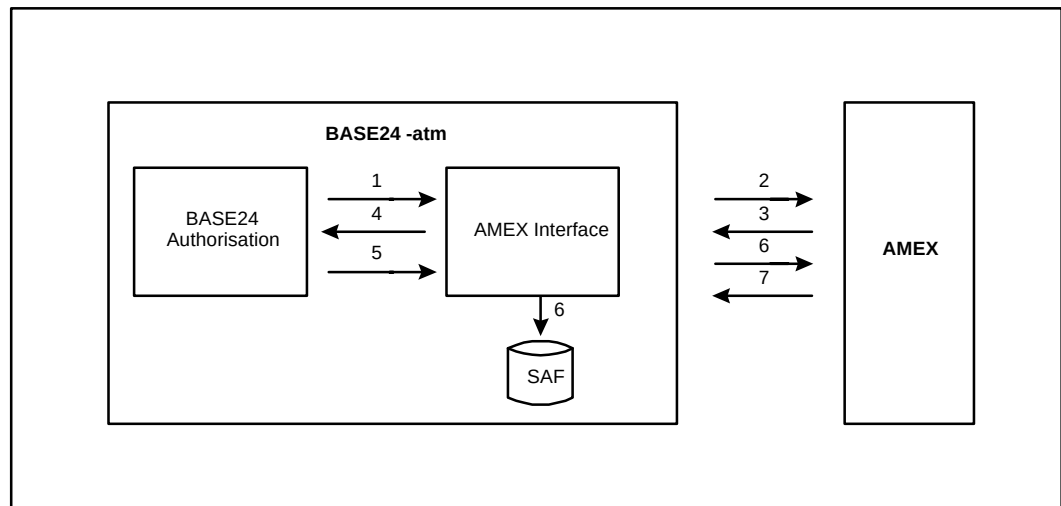


Figure 4-2 Financial Transaction Reversal Request/Response Message

A 1420 Reversal Request is sent to AMEX where:

- A communications or other outage has caused a time-out during a 1200 Financial Transaction Request.
 - The 1210 Request Response was received and was authorised and either a partial or non-dispense occurred.
 - The 1210 Request Response was not received within the ATM time-out period.
1. The BASE24 Authorisation formats a reversal request message (1420) to the BASE24-atm AMEX Interface, indicating that the transaction failed to complete.
 2. The BASE24-atm AMEX Interface formats an 1420 external message and sends the external message to AMEX for processing.
 3. AMEX sends an 1430 reversal response message to the BASE24-atm AMEX Interface. If a 1430 reversal response is not received after 30 seconds, the BASE24-atm AMEX Interface re-sends the 1420 message up to two times before discarding it.

Note:

Since authorisations are not recovered, but reversals are, it is possible that a 1420 reversal request message may arrive at AMEX where an 1210 authorisation request has been previously received. This sequence of events is handled by AMEX, who will reply with the appropriate response code.

4.2 NETWORK MANAGEMENT MESSAGE FLOWS

This section describes the message flows for the following .

- Sign-on/Sign-off Request Network Management Message
- Sign-on/Sign-off Response Network Management Message

4.2.1 Sign-on/Sign-off Request Messages (1804)

4.2.1.1 AMEX Initiated Sign-on

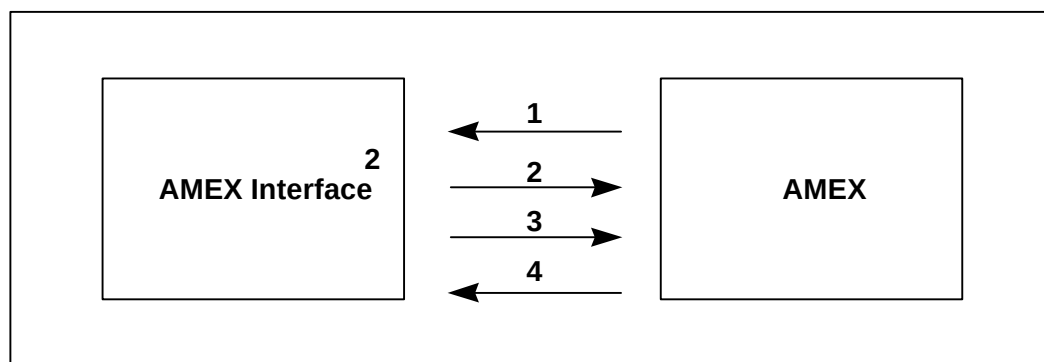


Figure 4-3 AMEX Initiated Sign-on

1. AMEX sends an 1804 sign-on request message to re-establish communications. The sign-on message is identified by the value 801 in the Function Code (P-24) field of the BASE24 external message.
2. The BASE24-atm AMEX Interface receives the message and performs the following steps:
 - a) Marks the BASE24-atm AMEX Interface as signed on.
 - b) Sends a sign-on response to AMEX. The sign-on response contains the appropriate action code in field 39 BASE24 external message to indicate:

800	Action accepted
904	Format error
912	Card issuer unavailable.
3. The BASE24-atm AMEX Interface sends an echo test request message containing the value 831 in the Function Code (P-24) field of the BASE24 external message.
4. If the connection between BASE24 and AMEX is available, AMEX sends an echo test response. The echo test response message is identified by the value 831 in the Function Code (P-24) field of the BASE24 external message.

4.2.1.2 AMEX Initiated Sign-off

1. AMEX sends an 1804 sign-off request message when the connection with the AMEX host goes down. The sign-off message is identified by the value 802 in the Function Code (P-24) field of the BASE24 external message.
2. If AMEX issue a manual sign-off, the link to the AMEX host will be dropped, followed by the link to the BASE24 system.

4.2.1.3 BASE24 Initiated Sign-off

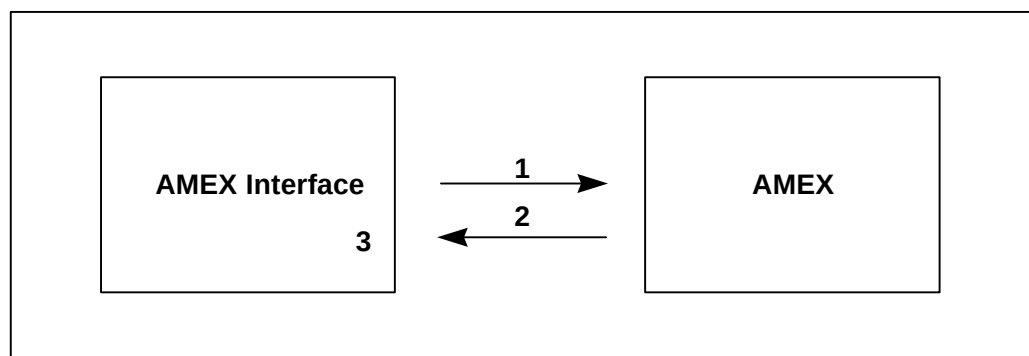


Figure 4-4 BASE24 Initiated Sign-off

1. The BASE24-atm AMEX Interface sends a sign-off request message to AMEX. The sign-off message is identified by the value 802 in the Function Code (P-24) field of the BASE24 external message.
2. AMEX does not reply to the sign-off message, but leaves the link up. AMEX will send repeated sign-on messages (every five minutes) until the link is back.
3. If AMEX receives a transaction request (1200) from BASE24 whilst BASE24 is signed-off, the transaction is accepted for processing and AMEX will take the transaction as an indicator that the link is open.

4.2.2 Sign-on/Sign-off Response Messages (1814)

This message will be sent by either AMEX or BASE24 in response to a sign-off or dialogue test message and will contain the appropriate action code in field 39 of the external message to indicate:

800	Action accepted
904	Format error
912	Card issuer unavailable.

5. ENCRYPTION FOR ATM TRANSACTIONS

AMEX uses hardware based DES encryption with a PIN key encrypted under a Zone Control Master Key (ZMCK). The encryption device is an IBM 4702.

Only PIN Block encryption is supported, there is no MACing.

5.1 TYPE OF ENCRYPTION

AMEX support IBM 3624 PIN block format across the link. The format is:

P,P,P,P,F,F,F,F,F,F,F,F,F,F

left-justified, pad character F.

For example:

If PIN = 1234, plain text PIN block = 1234 FFFF FFFF FFFF

All AMEX PINs are four digits in length.

5.2 ENCRYPTION KEYS

Two encryption keys are used:

- PIN Key
- Zone Control Master Key (ZCMK).

The PIN Key is used to encrypt the PIN block and is stored (encrypted under an HSM Key) on the KEYF in BASE24. (See section 3.5).

The ZCMK is generated in three parts. The key parts are XORed together to form the ZCMK.

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A. RESPONSE CODE TRANSLATION

AMEX RESPONSE CODE	BASE24-atm RESPONSE CODE	
000	000	Approved
100	068	Transaction denied
101	051	Card expired
104	068	Operation not permitted for card
106	062	PIN tries exceeded
110	074	Amount invalid
111	052	Invalid card number
117	053	Incorrect PIN
126	070	PIN block invalid
190	061	Limit exceeded
290	150	Refused - retain card
904	074	Incorrect message format
909	070	System error

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B. MESSAGE FORMATS

This appendix details the layout of the AMEX messages.

The abbreviations used to specify the presence of the data elements are listed in the table below.

ABBREVIATION	DETAILS
M	Mandatory data element for ISO.
MP	Mandatory data element for ISO for partial reversal requests.
MA	Mandatory data element for AMEX.

B.1 AUTHORISATION REQUEST (1200)

Below is the definition of the ISO message format for an 1200 ATM authorisation request.

BIT MAP POSITION	DATA ELEMENT	USAGE	COMMENTS
P-3	Processing Code	M	Always 014000 - cash request
P-4	Amount, Transaction	M	Transaction amount in ATM currency; format dependant on local currency
P-11	System Trace Audit Number	M	Unique transaction number
P-12	Date and Time, Local Transaction	M	Date and time of transmission to AMEX. For example, 980629165748 is 29 June 1998 at 16:57:48.
P-22	Point of Service Data Code	M	Not checked by AMEX
P-24	Function Code	M	Specific purpose of message. Valid values are: • 200 original authorisation
P-28	Reconciliation Date	MA	Format = YYMMDD
P-32	Acquiring Institution Identification Code	M	ISO Institution ID, identifying the acquiring institution
P-35	Track-2 Data	MA	The track 2 details from the AMEX card
P-37	Retrieval Reference Number	MA	Transaction sequence number
P-41	Card Acceptor Terminal Identification	MA	A unique number referencing the terminal, either the logical address supplied by BASE24 or supplied by the terminal manufacturer
P-43	Card Acceptor Name & Address	MA	Location of the terminal
P-49	Currency Code Transaction	MA	Local currency of acquirer or source location of the transaction, specifying the currency used in P-4.
P-52	PIN data	MA	Encrypted PIN block
P-58	Authorising Agent ID	MA	ISO Institution ID, identifying the acquiring institution

B.2 AUTHORISATION RESPONSE (1210)

Below is the definition of the ISO message format for an 1210 ATM authorisation response.

BIT MAP POSITION	DATA ELEMENT	USAGE	COMMENTS
P-4	Amount, Transaction	M	The same as sent in the 1200 request message
P-11	System Trace Audit Number	M	Unique transaction number. The same as sent in the 1200 request message
P-12	Date and Time, Local Transaction	M	Date and time of transmission to AMEX. The same as sent in the 1200 request message
P-22	Point of Service Data Code	M	As sent in the 1200 request message. Not checked by AMEX
P-24	Function Code	M	As sent in the 1200 request message
P-28	Reconciliation Date	MA	As sent in the 1200 request message
P-32	Acquiring Institution Identification Code	M	As sent in the 1200 request message
P-35	Track-2 Data	M	As sent in the 1200 request message
P-39	Action Code	M	Response from the authorisation
P-41	Card Acceptor Terminal Identification	MA	As sent in the 1200 request message
P-49	Currency Code, Transaction	MA	As sent in the 1200 request message
P-58	Authorising Agent ID	MA	As sent in the 1200 request message

B.3 REVERSAL REQUEST MESSAGE (1420)

Below is the definition of the ISO message format for an 1420 ATM reversal request message.

BIT MAP POSITION	DATA ELEMENT	USAGE	COMMENTS
P-2	Primary Account Number (PAN)	M	Card number. AMEX cards always have 15 digits
P-4	Amount, Transaction	M	Transaction amount in ATM currency; format dependant on local currency
P-11	System Trace Audit Number	M	Unique transaction number
P-12	Date and Time, Local Transaction	M	Date and time of transaction
P-24	Function Code	M	Indicates purpose of message: 400 - Full reversal, transaction did not complete as approved 401 - Partial reversal, transaction did not complete for full amount
P-25	Message Reason Code	M	Reason for reversal
P-28	Reconciliation Date	MA	Format = YYMMDD
P-30	Amounts, Original	MP	Contains a copy of the Amount, Transaction field from the corresponding 1200 authorisation request message
P-32	Acquiring Institution Identification Code	M	ISO Institution ID, identifying the acquiring institution
P-41	Terminal ID	MA	As sent in the 1200 request message
P-49	Currency Code Transaction	MA	As sent in the 1200 request message
P-56	Original Data Elements	M	Copies of data elements contained in original 1200 request message
P-58	Authorising Agent ID	MA	ISO Institution ID, identifying the acquiring institution

B.4 REVERSAL RESPONSE MESSAGE (1430)

Below is the definition of the ISO message format for an 1430 ATM reversal response message.

BIT MAP POSITION	DATA ELEMENT	USAGE	COMMENTS
P-2	Primary Account Number (PAN)	M	Card number. Same as sent in 1420 message
P-4	Amount, Transaction	M	As sent in 1420 message
P-11	System Trace Audit Number	M	As sent in 1420 message
P-12	Date and Time, Local Transaction	M	As sent in 1420 message
P-28	Reconciliation Date	MA	Format = YYMMDD
P-30	Amounts, Original	MA	Contains a copy of the Amount, Transaction field from the corresponding 1200 authorisation request message
P-32	Acquiring Institution Identification Code	M	As sent in 1420 message
P-39	Action Code	M	Valid values: 400 - reversal accepted 480 - ignored, timer expired 481 - ignored, other reasons
P-49	Currency Code Transaction	M	As sent in 1420 message
P-58	Authorising Agent ID	MA	As sent in 1420 message

B.5 NETWORK MANAGEMENT REQUEST (1804)

Below is the definition of the ISO message format for an 1804 network management request.

BIT MAP POSITION	DATA ELEMENT	USAGE	COMMENTS
P-1	Bit Map, Secondary	M	
P-11	System Trace Audit Number	M	Unique transaction number
P-12	Date and Time, Local Transaction	M	Date and time of local transaction
P-24	Function Code	M	Purpose of message. Valid values: 801 - Sign-on 802 - Sign-off 831 - Echo Test
S-93	Transaction Destination Institution Identification Code	M	Identifier of destination institution
S-94	Transaction Originator Institution Identification Code	M	Identifier of originating institution

B.6 NETWORK MANAGEMENT RESPONSE (1814)

Below is the definition of the ISO message format for an 1814 network management response.

BIT MAP POSITION	DATA ELEMENT	USAGE	COMMENTS
P-1	Bit Map, Secondary	M	
P-11	System Trace Audit Number	M	Unique transaction number. As sent in the 1804 request message
P-12	Date and Time, Local Transaction	M	As sent in the 1804 request message
P-39	Action Code	M	Action performed by 1804 message. Valid values are: 800 - Accepted (applies to all 1804 messages) 904 - Format error (applies to all 1804 messages) 912 - Card Issuer Unavailable (1804/Sign-on only)
S-93	Transaction Destination Institution Identification Code	M	Identifier of destination institution
S-94	Transaction Originator Institution Identification Code	M	Identifier of originating institution

C. LOG MESSAGES

Log messages are written to the BASE24 Disc Log File and, optionally, to other logging devices defined by the BASE24 log configuration.

Additional log messages are displayed when diagnostic tracing is turned on in a process by means of the NCS TRACE command. These messages have not been documented in this manual, as they are self-explanatory. They can be identified by the prefix 'T -' in the message text.

C.1 FORMAT OF LOG MESSAGES

Log messages are generated to report an error condition or information. Error messages may require corrective action before the BASE24-atm EMEA AMEX Interface can function properly.

The standard format for BASE24 log messages is:

YY/MM/DD HH:MM:SS (X) <PROCESS ID> <OBJECT> MMMM RRRR <TEXT>

where: 'YY/MM/DD' specifies the date the message was sent.

'HH:MM:SS' specifies the time the message was sent.

'(X)' specifies the severity code where:

(0) or (I)	Status (Informational)
(1) or (W)	Warning
(2) or (E)	Error
(3) or (C)	Critical

'<PROCESS ID>' specifies the symbolic process name.

'<OBJECT>' specifies a 16-byte string defined and passed by the process.

'MMMM' specifies the message number.

'RRRR' specifies the routing code.

'<TEXT>' specifies the text of the message.

For example:

**98/02/08 12:00:00 (2) AMEX 0422 0002 CARD ISSUER UNAVAILABLE
- LINK MARKED DOWN**

To avoid the repetition of showing the date, time, process identifier, object and routing code for each message; only the message number, severity code and the text of the message are documented. Therefore, the display format of log messages adopted for documentation is:

MMMM (X) <TEXT>

For example:

0422 (2) CARD ISSUER UNAVAILABLE - LINK MARKED DOWN

For each log message, additional information will be given:

- the cause or reason the message is displayed.
- the remedy or solution for the message, if relevant.

Also, the following support/service personnel may need to be contacted:

- *Operations Personnel*

In-house operations staff who are responsible for initiating fault action for a particular problem.

- *HELP24*

ACI Limited personnel who are responsible for solving customer queries and problems.

C.2 ATM LOG MESSAGES

The following are log messages generated specifically by the BASE24-atm EMEA AMEX Interface. For details of the standard HISO messages, refer to the BASE24 ISO Host Interface Manual.

0422 0426 0428	(2)	CARD ISSUER UNAVAILABLE - LINK MARKED DOWN
---	------------	---

CAUSE: Unable to connect to the AMEX system.

REMEDY: If the problem persists, contact AMEX.

0425 0427	(1)	NMM REQUEST DENIED - ACTION CODE = \\
----------------------------	------------	--

where: '\\ ' specifies the action code.

CAUSE: NMM request from AMEX denied.

REMEDY: If the problem persists, contact AMEX.

0532	(0)	DPC ##### SAF RETRIES DISABLED - SAF RECORD FOLLOWS: \?
-------------	------------	--

where: '#####' specifies the DPC number.

'\?' specifies the SAF record.

CAUSE: No SAF retries permitted by parameter on HCF.

REMEDY: Contact Operations personnel.

0971	(0)	INVALID MESSAGE FROM ISSUER DROPPED - MESSAGE: \?
-------------	------------	--

where: '\?' specifies the message number.

CAUSE: An unrecognised message was received from AMEX.

REMEDY: Information only, no action required. If problem persists, contact AMEX.

1349	(0)	\? param invalid in the LCONF
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where: '\?' specifies the parameter.

CAUSE: The ATM-AMEX-ISSUER-INST-ID parameter has not been set up correctly in the LCONF.

REMEDY: Contact Operations personnel to ensure this parameter is set correctly.

1351	(2)	\? param not found in the LCONF
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where: '\?' specifies the parameter.

CAUSE: A parameter has not been set up correctly or is missing from the LCONF.

REMEDY: Contact Operations personnel to ensure this parameter is added to the LCONF or, if present, is set correctly.

1352	(0)	\? param not found in the LCONF
where:	'?' specifies the parameter.	
CAUSE:	A parameter has not been set up correctly or is missing from the LCONF.	
REMEDY:	Contact Operations personnel to ensure this parameter is added to the LCONF or, if present, is set correctly.	
1360	(2)	LCONF param \S; Too many INST IDs assigned
where:	'S' specifies the LCONF parameter.	
CAUSE:	There are too many ATM-AMEX-ACQ-INST-ID-nn parameters are configured in the LCONF.	
REMEDY:	Remove some of the parameters from the LCONF.	
1361		
1362		
1364	(1)	LCONF param \S; Unable to validate Acquirer INST ID \?
where:	'S' specifies the LCONF parameter.	
	'?' specifies the Institution Identifier.	
CAUSE:	The Acquirer Institution ID in the LCONF ATM-AMEX-ACQ-INST-ID-nn parameter is invalid.	
REMEDY:	Correct the Institution ID.	
1363		
1365	(1)	LCONF param \?; Term FIID \?; Incorrect
where:	'?' specifies the LCONF parameter.	
	'?' specifies the terminal FIID.	
CAUSE:	The Terminal FIID in the LCONF ATM-AMEX-ACQ-INST-ID-nn parameter is invalid.	
REMEDY:	Correct the FIID.	
1366	(1)	Error \e on LCONF file \f for entry \?
where:	'e' specifies the error number.	
	'f' specifies the LCONF file name.	
	'?' specifies the LCONF key	
CAUSE:	Error on reading the LCONF.	
REMEDY:	Contact Operations personnel.	
1367	(1)	LCONF param \?; No INST IDs defined in LCONF
where:	'?' specifies the LCONF parameter.	
CAUSE:	There are no ATM-AMEX-ACQ-INST-ID-nn parameters configured in the LCONF.	
REMEDY:	Information only, no action required.	



USER'S COMMENTS

BASE24-atm EMEA AMEX INTERFACE MANUAL

Release 5.0

Please assist us in our future documentation plans by sending us your comments. We suggest that you use the form below, but please feel free to supplement it with more detailed comment as necessary.

NAME :

DATE :

COMPANY :

ADDRESS :

A. PERSONAL EXPERIENCE

1. How long have you been using the product?

0-3 months

3-9 months

Over 9 months

☐☐☐

2. In what capacity do you use the product?

Project Management

Development

Operations/Network

Test

☐☐☐☐

B. THIS DOCUMENT

1. Have you had any difficulty with the product where you feel that this document ought to have helped but failed to do so? If yes, please give details:

2. In particular, we would appreciate your comments on the following aspects of the document:

		YES	NO
CONTENTS/INDEX	Is it helpful and is the level of detail sufficient?	☐	☐
STRUCTURE	Is the structure of the document satisfactory?	☐	☐
USEFULNESS	Do you often use this document in your day-to-day work?	☐	☐

3. Have you found the document useful? If so, please tell us what you particularly like about it:

4. Please give examples of anything that you have found to be unsatisfactory and suggest any improvements that could be made:

C. CURRENT DOCUMENTATION

	YES	NO
1. Does the current range of manuals/user's guides cover the areas of system use on which you require information?	☐	☐
2. If coverage is inadequate, please suggest what other areas could be covered:		

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